EXPERIMENT 10

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AIM:

The aim is to create data visualizations, such as pie charts and bar graphs, for an inventory management system using JavaScript.

PROCEDURE:

Step 1:

Set Up Your HTML File

Welcome INDEX.HTML

INDEX.HTML > html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <title>Student Performance Dashboard</title>
    <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
    <style>
        body {
            font-family: Arial;
            background: #f4f4f4;
            text-align: center;
        }
        h1 {
            margin-top: 20px;
        }
        .container {
            width: 80%;
            margin: auto;
        }
        canvas {
            margin: 30px;
            background: white;
            padding: 20px;
            border-radius: 10px;
        }
    </style>
</head>
<body>

<h1>Student Performance Analysis</h1>

<div class="container">
    <canvas id="barChart"></canvas>
    <canvas id="doughnutChart"></canvas>
</div>

<script>
```

Step 2:

Create the JavaScript File for Charts Next, create a JavaScript file (script.js) to handle the data visualization logic.

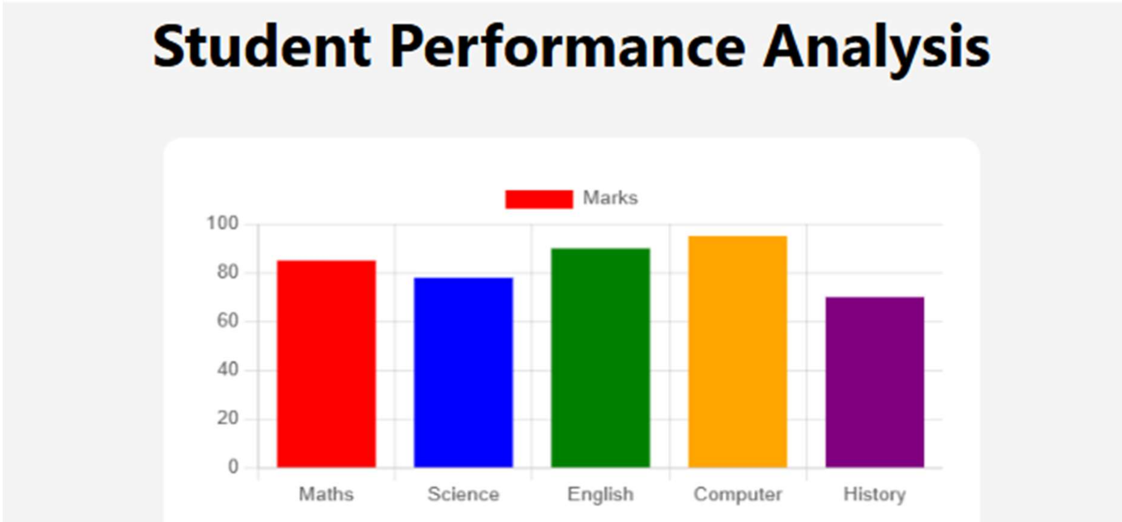
```
Welcome    INDEX.HTML
INDEX.HTML > html
<html lang="en">
<body>
<script>
    // Bar Chart - Subject Marks
    const barCtx = document.getElementById('barChart').getContext('2d');

    new Chart(barCtx, {
        type: 'bar',
        data: {
            labels: ['Maths', 'Science', 'English', 'Computer', 'History'],
            datasets: [{
                label: 'Marks',
                data: [85, 78, 90, 95, 70],
                backgroundColor: ['red', 'blue', 'green', 'orange', 'purple']
            }]
        },
        options: {
            responsive: true,
            plugins: {
                legend: {
                    display: true
                }
            }
        }
    });

    // Doughnut Chart - Grade Distribution
    const doughnutCtx = document.getElementById('doughnutChart').getContext('2d');

    new Chart(doughnutCtx, {
        type: 'doughnut',
        data: {
            labels: ['A Grade', 'B Grade', 'C Grade', 'D Grade'],
            datasets: [{
                data: [40, 30, 20, 10],
                backgroundColor: ['green', 'blue', 'orange', 'red']
            }]
```

Welcome INDEX.HTML

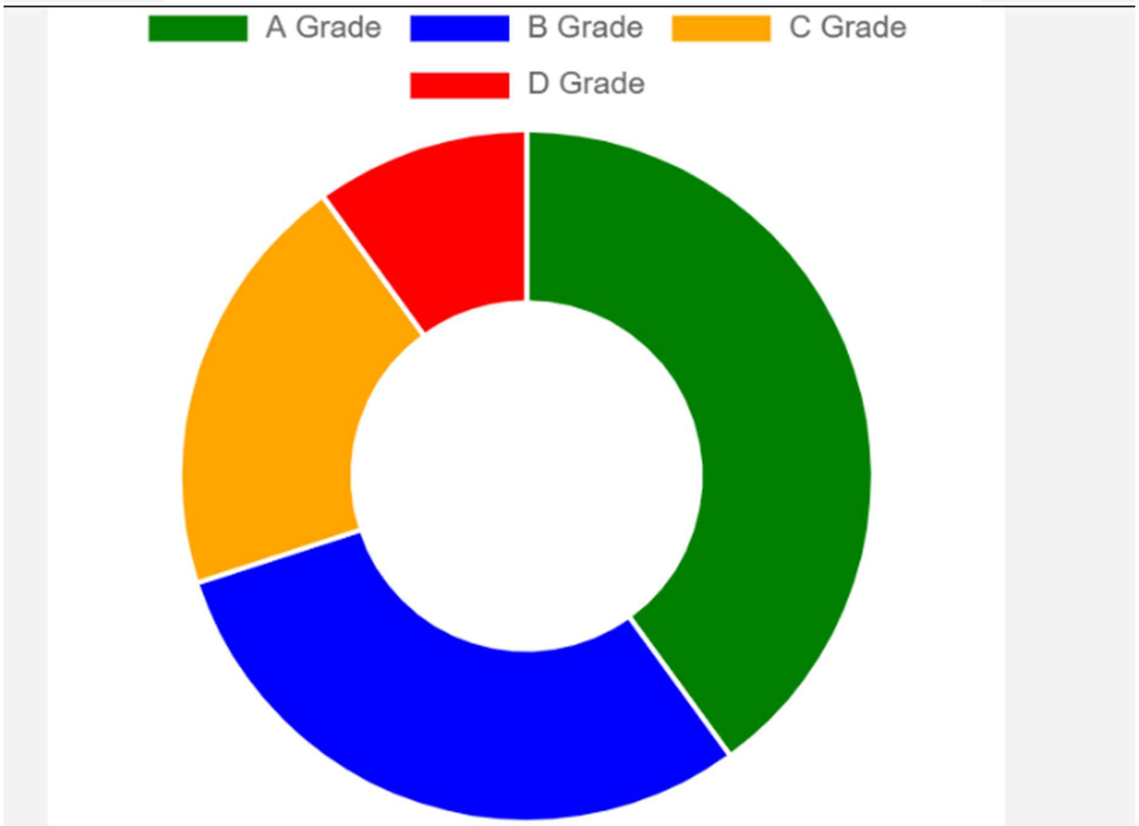
INDEX.HTML > html

```html
<html lang="en">
<body>
<script>
        options: {
            plugins: {
            }
        }
    });

    // Doughnut Chart - Grade Distribution
    const doughnutCtx = document.getElementById('doughnutChart').getContext('2d');

    new Chart(doughnutCtx, {
        type: 'doughnut',
        data: {
            labels: ['A Grade', 'B Grade', 'C Grade', 'D Grade'],
            datasets: [{
                data: [40, 30, 20, 10],
                backgroundColor: ['green', 'blue', 'orange', 'red']
            }]
        },
        options: {
            responsive: true
        }
    });
</script>

</body>
</html>
```

OUTPUT:



Student Performance Analysis
Marks
100
80
60
40
20
0
Maths
Science
English
Computer
History



A Grade
B Grade
C Grade
D Grade